Risk Management in Financial Markets

Pricing a principal protected structured note and evaluating a U.S. value equity fund

Cédric Senah | Academic case study, Master in Finance (2025)

About this project

This is a school project from my Master in Finance. I play the role of an analyst in a bank (a made up bank called **Winston Capital**) that wants to sell two new products to its clients. My job is to check if these products are a good idea, how much they cost, how much the bank earns and what can go wrong.

- **Part A, a safe investment linked to the stock market.** The client puts in USD 1,000 for one year. If the U.S. stock market (S&P 500) goes up, the client gets a part of the gain. If it goes down, the client still gets the USD 1,000 back. I explain how the bank builds this product and how much it earns on it.
- **Part B, a fund that buys cheap stocks.** The fund buys the U.S. companies that look the cheapest compared to the value in their accounts ("value" stocks). I test how this strategy performed in the past, whether clients would want it, and how much money the fund could lose in a bad month.

The case description was rewritten in my own words, and a few figures were corrected after the original submission. Tools used: Python and Excel.

Repository contents

File	Content
report.pdf	This report
structured_note_pricing.py	Option pricing and bank margin (Part A)
value_fund_analysis.xlsx	Performance ratios and VaR of the value strategy (Part B)

Data sources: S&P 500 daily levels (Investing.com), 13 week U.S. Treasury bill yield (Yahoo Finance, ticker ^IRX), Kenneth R. French Data Library ("Portfolios Formed on Book to Market" and "Fama/French 3 Factors").

Part A. The principal protected structured note

Case summary

The bank wants to sell a one year product at USD 1,000. At the end of the year:

- if the S&P 500 went up, the client gets USD 1,000 plus 25% of the gain (for example, +20% for the index gives 1,000 + 25% × 20% × 1,000 = USD 1,050);
- if the S&P 500 went down, the client simply gets USD 1,000 back.

Questions: how can the bank build this product, what is it worth, how much does the bank earn, and what are the risks?

A.1 Product structure

The payoff to the investor is $1{,}000 + 0.25 \times 1{,}000 \times \max(R_{S\&P}, 0)$. It can be replicated with two instruments that the bank buys with the USD 1,000 collected:

- **A long position in a one year zero coupon U.S. Treasury bond** with a face value of USD 1,000. Bought at a discount today, it is worth exactly USD 1,000 at maturity, whatever the equity market does. This is the mechanism that guarantees the principal.
- **A long position in 0.25 European at the money call option** on the S&P 500 (strike equal to the index level on the launch date, maturity one year). The call pays nothing if the index falls and pays 25% of the index gain if it rises, which delivers the equity participation.

From the bank's point of view, it is **short the note** (a liability to its clients) and **long the bond and the fractional call** as a hedge. Whatever is left from the USD 1,000 after buying both components is the bank's margin.

A.2 Value of the embedded option

Model. The option is a European call, so it is priced with the Black Scholes Merton model with a continuous dividend yield:

$C = S\, e^{-qT}\, N(d_1) - K\, e^{-rT}\, N(d_2)$, with $d_1 = [\ln(S/K) + (r - q + \sigma^2/2)\, T] / (\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$.

Following the case guidance, the option is written on an artificial asset worth USD 1,000 with a strike of USD 1,000 (at the money) and the volatility of the S&P 500, so that its price is directly expressed per note.

Python code. I used Python (Google Colab) to compute the volatility of the S&P 500 from daily prices and to download the risk free rate (13 week U.S. Treasury bill, ticker ^IRX) from Yahoo Finance.

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import yfinance as yf
from google.colab import drive


# Définir le chemin du fichier dans Google Drive
file_path = '/content/drive/MyDrive/S&P 500 Historical Data.csv'  # Modifier selon votre chemin
df = pd.read_csv(file_path, parse_dates=['Date'], dayfirst=True)

# Trier les données par date
df.sort_values(by='Date', ascending=True, inplace=True)
```

```python
# Convertir les prix en float et calculer les rendements
df['Price'] = df['Price'].replace(',', '', regex=True).astype(float)
df['Returns'] = df['Price'].pct_change()

# Calcul de la volatilité annualisée (252 jours de trading par an)
rolling_window = 252
df['Volatility'] = df['Returns'].rolling(window=rolling_window).std() * np.sqrt(252)
```

```python
# Télécharger les taux sans risque (1Y Treasury Yield) depuis Yahoo Finance
ticker = '^IRX'  # 13-week T-Bill comme approximation
treasury_data = yf.download(ticker, period='1y', interval='1d')
rf_rate = treasury_data['Close'] / 100  # Convertir en taux décimal
```

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```

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# Vérifier les colonnes disponibles
print("Colonnes disponibles dans treasury_data:", treasury_data.columns)

# Aplatir les données en supprimant le MultiIndex
treasury_data.columns = treasury_data.columns.droplevel(1)

# Vérifier si 'Adj Close' existe, sinon utiliser 'Close'
if 'Adj Close' in treasury_data.columns:
    rf_rate = treasury_data[['Adj Close']].copy()
elif 'Close' in treasury_data.columns:
    rf_rate = treasury_data[['Close']].copy()
else:
    raise KeyError("Ni 'Adj Close' ni 'Close' ne sont disponibles dans les données du Treasury Yield")
```

Figure 1. Loading the S&P; 500 data, computing returns and volatility, downloading the risk free rate

```python
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else:
    raise KeyError("Ni 'Adj Close' ni 'Close' ne sont disponibles dans les données du Treasury Yield")

# Convertir l'index de rf_rate en datetime et normaliser les dates
rf_rate.index = pd.to_datetime(rf_rate.index)
rf_rate = rf_rate / 100  # Convertir en taux décimal

# Réinitialiser l'index pour fusionner correctement
rf_rate.reset_index(inplace=True)
rf_rate.rename(columns={'index': 'Date'}, inplace=True)

# Fusionner les taux sans risque avec les données du S&P 500
df = df.merge(rf_rate, on='Date', how='left')
df.rename(columns={'Adj Close': 'Risk-Free Rate', 'Close': 'Risk-Free Rate'}, inplace=True)

# Afficher les résultats
plt.figure(figsize=(12,5))
plt.plot(df['Date'], df['Volatility'], label='Volatilité Annualisée')
plt.xlabel('Date')
plt.ylabel('Volatilité')
plt.title('Volatilité Annualisée du S&P 500')
plt.legend()
plt.show()

plt.figure(figsize=(12,5))
plt.plot(df['Date'], df['Risk-Free Rate'], label='Risk-Free Rate', color='orange')
plt.xlabel('Date')
plt.ylabel('Taux sans risque')
plt.title('Évolution du Taux sans Risque (US 1Y Treasury)')
plt.legend()
plt.show()

# Afficher un aperçu des données
print(df[['Date', 'Volatility', 'Risk-Free Rate']].dropna().tail(10))
```

Figure 2. Merging the two series and plotting the volatility and the risk free rate

```
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Colonnes disponibles dans treasury_data: MultiIndex([( 'Close', '^IRX'),
            (  'High', '^IRX'),
            (   'Low', '^IRX'),
            (  'Open', '^IRX'),
            ('Volume', '^IRX')],
           names=['Price', 'Ticker'])
```

	Date	Volatility	Risk-Free Rate
2709	2024-12-17	4.868811	0.04240
2717	2024-12-18	4.881151	0.04232
2725	2024-12-19	4.864018	0.04220
2733	2024-12-20	4.888277	0.04215
2757	2024-12-23	4.904980	0.04215
2763	2024-12-24	4.936520	0.04200
2769	2024-12-26	5.009373	0.04215
2777	2024-12-27	4.962749	0.04178
2801	2024-12-30	4.937777	0.04182
2808	2024-12-31	4.938109	0.04208

Figure 3. Output of the code for the last days of December 2024

Input	Value	Justification
Underlying S / strike K	1,000 / 1,000	At the money, scaled to the note's principal
Maturity T	1 year	1 March 2025 to 1 March 2026
Risk free rate	4.20% (4.11% cont.)	13 week T bill yield at end 2024, converted: ln(1.042)
Dividend yield q	1.26%	S&P 500 dividend yield at the valuation date
Volatility σ	13.08%	1 year annualised vol. of daily S&P 500 returns

Results.

Output	Value
d1 / d2	0.284 / 0.153
Call delta	0.604
Price of one ATM call	USD 65.88
Embedded option: 0.25 × 65.88	USD 16.47

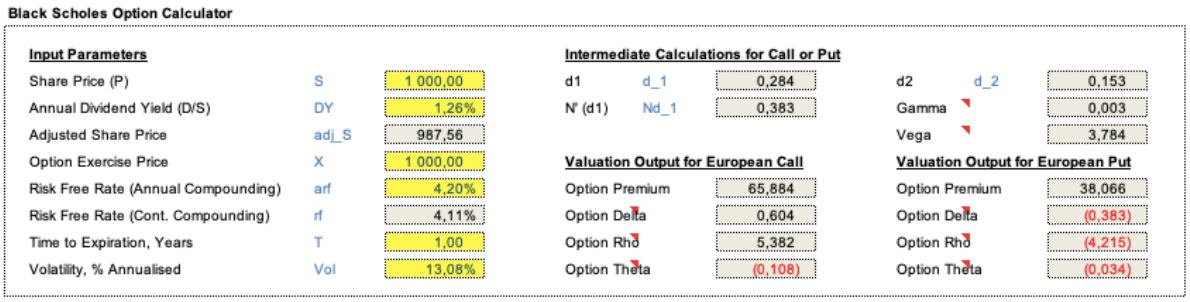


Figure 4. Black Scholes calculator used to price the at the money call

The price was obtained with a Black Scholes spreadsheet calculator and cross checked with the Python script *structured_note_pricing.py* included in the repository (USD 65.86, the small gap comes from rounding of the inputs). Since the note only pays 25% of the index return, it embeds a quarter of a call, worth **USD 16.47 per note**.

Note: an earlier version of the volatility script returned values around 4.9, which reflects a units error in how returns were computed from the raw price column. The volatility used in the pricing (13.08%) is consistent with the realised S&P 500 volatility in 2024, and the published script computes it as a decimal.

A.3 Margin for Winston Capital

Component	USD per note
Selling price of the note	1,000.00
Zero coupon bond: 1,000 / 1.042	959.69
Embedded option: 0.25 call	16.47
Total replication cost	976.16
Gross margin	23.84 (2.38%)

The bank earns about **USD 23.84 per note, or 2.38% upfront**, for a one year product. This gross margin must still cover distribution, structuring, legal and hedging costs (bid ask spreads, rebalancing), so the net margin will be lower. It is in line with typical upfront fees on retail structured products, which makes the product worth offering.

A useful benchmark: if the bank kept no margin at all, it could offer a participation of (1,000 − 959.69) / 65.88 ≈ **61%**. Offering 25% is therefore a choice to keep most of the option budget as profit, which is commercially viable when rates are high (cheap bond) but may look unattractive to well informed investors. A more competitive participation (for example 40% to 45%) would still leave a margin of roughly 1% to 1.4%.

A.4 Risks for the bank

- **Pricing and market risk before launch.** The note is priced on 20 February but hedged on 1 March. If volatility rises, the call becomes more expensive; if the index moves, the at the money strike is reset at a different level.
- **Interest rate risk.** If rates *fall* before the hedge is put in place, the zero coupon bond becomes more expensive, which eats into the margin (rising rates have the opposite, favourable effect).
- **Hedging and model risk.** If the bank does not buy the option but replicates it through delta hedging, its result depends on realised volatility, jumps and transaction costs, which Black Scholes does not capture.
- **Counterparty risk.** An OTC option bought from another dealer may not be paid if that dealer defaults, while the bank still owes the coupon to its clients.
- **Liquidity risk.** In a stressed market, unwinding the hedge or buying back notes from clients may only be possible at a large discount.
- **Dividend risk.** Lower than expected dividends raise the forward and the value of the call.
- **Legal, regulatory and reputational risk.** Retail structured products are subject to suitability and disclosure rules; mis selling complaints or a perceived excessive margin may damage the bank's reputation.

A.5 Risk mitigation

- **Lock the hedge at launch**: buy the bond and the option on the issue date, or fix the final participation rate at launch ("indicative 25%, final terms on 1 March"), so that market moves during the subscription period are passed on.
- **Hedge rate exposure** during the subscription period with Treasury futures or forward rate agreements.
- **Static rather than dynamic hedge**: buying the listed or OTC call back to back removes model risk; if delta hedging is used, set vega and gamma limits and monitor them daily.
- **Counterparty controls**: trade under ISDA/CSA agreements with collateral, prefer exchange traded options where possible and diversify dealers.
- **Liquidity management**: define a buy back policy (secondary market with a bid ask spread) and size the issuance to the depth of the hedging market.
- **Compliance**: clear key information documents, target market definition and suitability checks.

Part B. The U.S. Value Fund

Case summary

The bank wants to open a fund that buys the 10% of U.S. stocks that are the cheapest compared to their book value (their value in the accounts), and changes the list every month. Using real historical data from the Kenneth R. French website, I check whether this strategy did well in the past, how much the bank could charge for it, why it could keep working, and how much a USD 300 million fund could lose in one month (Value at Risk).

B.1 Historical performance

Indicator	Value (monthly)	Meaning
Average return	1.32%	About 17% per year compounded
Volatility	9.06%	About 31% annualised, much higher than the market
Beta vs market	0.92	Sensitivity to Mkt−RF
Sharpe ratio	0.116	Excess return per unit of total risk (about 0.40 annualised)
Sortino ratio	0.169	Excess return per unit of downside risk

Methods. Sharpe = $(R_p - R_f) / \sigma_p$; Sortino uses only the standard deviation of negative excess returns; beta is the slope of the regression of the portfolio's excess return on the market's excess return; Jensen's alpha = $R_p - [R_f + \beta (R_m - R_f)]$.

Jensen's alpha. Using the sample averages ($R_f \approx 0.27\%$ and Mkt−RF ≈ 0.68% per month over the full Fama French sample), alpha ≈ 1.32% − 0.27% − 0.92 × 0.68% ≈ **0.42% per month**, about 5% per year. The original submission reported 1.07%, which corresponds to deducting only $\beta \times R_f$ instead of the full CAPM expected return; the corrected figure is computed in the spreadsheet.

The strategy earns a high average return and a positive CAPM alpha, but with a very high volatility and deep drawdowns (value stocks were hit hard in 1929 to 1932, 2008 to 2009 and 2020), so its risk adjusted return is close to the market's.

B.2 Attractiveness and management fee

The fund's monthly Sharpe ratio (0.116) is slightly below that of the market portfolio over the same sample (roughly 0.13), even though its raw return and alpha are higher. In other words, investors are paid for the extra return mainly by taking more risk. A rational investor would not pay active fees for a product that delivers a lower Sharpe ratio than a cheap index fund, unless they value the diversification of a value tilt in a broader portfolio.

A competitive positioning is therefore a **management fee of around 0.40% to 0.75% per year**: well above an S&P 500 ETF (below 0.10%) to reflect the alpha, but far below the 5% annual alpha so that most of the premium is left to the client, and in line with existing value factor funds. Fees above 1% would likely push investors towards cheaper smart beta ETFs.

B.3 Why the strategy could work in the future

- **Risk based explanation** (Fama and French): high book to market firms are often distressed or less flexible; their higher expected return is compensation for a risk that does not disappear and should persist.

- **Behavioural explanation** (Lakonishok, Shleifer and Vishny): investors extrapolate past bad news and overreact, pushing these stocks too low; prices mean revert as fundamentals turn out better than feared.
- **Limits to arbitrage**: career risk and the long periods of underperformance discourage professional investors from fully arbitraging the premium.
- **Long track record and international evidence** across many markets and decades.

Honesty with clients requires noting that value underperformed growth for most of 2010 to 2020; the premium is volatile and requires a long holding horizon. Combining value with quality or momentum can reduce this risk.

B.4 Parametric VaR (USD 300 million, one month)

VaR = μ − z × σ, with μ = 1.32% and σ = 9.06%, assuming normally distributed returns.

Confidence	z	VaR (%)	VaR (USD)
95%	1.645	−13.58%	−40.75 M
99%	2.33	−19.79%	−59.38 M

In one month out of twenty, the fund is expected to lose more than USD 40.75 million; in one month out of a hundred, more than USD 59.38 million. The normality assumption tends to understate the frequency of extreme losses.

B.5 Historical VaR at 95%

Method	VaR 95% (%)	VaR 95% (USD)	Basis
Parametric	−13.58%	−40.75 M	Normal distribution
Historical simulation	−11.95%	−35.84 M	5th percentile of actual monthly returns

The historical 95% VaR is lower than the parametric one. The empirical distribution is more peaked than the normal: many months are close to the mean, which pulls the 5th percentile closer to zero, while the variance is inflated by a few extreme months. This does not mean the fund is less risky: fat tails usually show up further out, so at 99% or with Expected Shortfall the historical measure would typically exceed the normal one. A prudent risk framework would report both measures and complement them with stress tests based on episodes such as 1932, 2008 and March 2020.